SKILLNEXIS WEEK 4 CAPSTONE

Sales Performance Analysis and Prediction

PDF Explanation Report

Submitted by

Maithili Shailesh Jadhav

Project objective

Analyze sales data, identify business trends, build a Linear Regression model to predict sales, and present insights through an interactive Power BI dashboard.

Tools used

Python, Pandas, NumPy, Matplotlib, scikit-learn, Jupyter Notebook, Microsoft Power BI, and CSV data files.

Dataset overview

The dataset contains 700 rows and 16 columns, including Segment, Country, Product, Units Sold, Sales, Profit, Date, and Year.

1. Data Preparation and Exploratory Analysis

Data cleaning

Extra spaces were removed from column names and the Date field was converted to datetime. The original dataset had 53 missing values in Discount Band; these were filled with “None” to retain records where a discount band was not specified. Duplicate rows were checked, and none were found. The cleaned dataset was saved for Power BI.

Key business metrics

Metric	Result
Total Sales	118,726,350.26
Total Profit	16,893,702.26
Total Units Sold	1,125,806
Average Sales per Record	169,609.07
Records	700

Findings from EDA

- **Country:** United States of America had the highest sales (about 25.03 million), followed by Canada (about 24.89 million).
- **Segment:** Government generated the highest sales (about 52.50 million), followed by Small Business (about 42.43 million).
- **Product:** Paseo was the highest-selling product by sales (about 33.01 million).
- **Year:** Sales in 2013 were about 26.42 million, while 2014 sales were about 92.31 million.

These comparisons identify which markets, segments, and products contributed most to recorded sales. The difference between years should be interpreted in the context of the periods and records represented in the dataset.

2. Sales Prediction Model

A Linear Regression model was trained to estimate Sales using Units Sold, Sale Price, Manufacturing Price, Month Number, and Year as input features. The dataset was split into training and testing sets using an 80:20 split with random_state=42. Sales was used as the target variable.

Evaluation results

Metric	Value	Meaning
MAE	83,139.20	Average absolute prediction error
MSE	13,051,282,466.61	Average squared prediction error
R^2 Score	0.7643	Explains about 76.43% of test-set variation

The R^2 result indicates that the model captures a substantial portion of variation in the test data, but predictions still have error. This is a baseline model; performance could be improved by testing additional relevant features, comparing algorithms, and using cross-validation.

3. Power BI Dashboard

The dashboard includes KPI cards for Total Sales, Total Profit, and Total Units Sold. Charts show Sales by Country, Sales by Segment, Sales by Product, Sales by Year, and Profit by Country. Slicers for Year, Segment, and Product let users filter the visuals interactively.

4. Recommendations

- Monitor the Government and Small Business segments because they contribute the largest sales totals.
- Review Paseo's performance and investigate opportunities to improve lower-performing products.
- Compare approaches across countries and investigate opportunities in lower-sales markets.
- Use dashboard filters to monitor sales and profit by year, segment, and product.
- Evaluate alternative regression models and review prediction errors across groups.

5. Conclusion

This capstone combines data cleaning, exploratory analysis, sales prediction, and interactive reporting. Python supports reproducible analysis and model evaluation, while Power BI makes business metrics easier to explore.

Included project files: Jupyter Notebook (.ipynb), cleaned dataset (.csv), Power BI report (.pbix), findings document (.docx), and this PDF explanation report.